

Fabric IQ in the Real World

Chris Hyde

Matt Gordon

Thank You To Our Sponsors For Making SQL Saturday Possible!



KEEP YOUR TAX DOLLARS LOCAL!

The PEACH Education Tax Credit lets you redirect your Georgia state income tax dollars directly to Innovation Academy — at no additional cost to you. This is a 100% dollar-for-dollar tax credit (not a deduction).

WHY THIS MATTERS

Instead of sending your tax dollars to the state's general fund, you can directly support:

- Mentorships & internships
- STEM and innovation programs
- Curriculum development
- Teacher professional growth
- Real-world student experiences

ELIGIBILITY LIMITS

- **Single Filer:** UP TO \$2,500
- **Filing Jointly:** Up To \$5,000
- **LLC Member or Pass Through Entity:** Up to \$25,000
- **C or S Corporation:** up to 75% of annual Georgia income tax liability

Same Taxes. Greater Impact.

STEP 1: APPLY

- Apply for the PEACH Tax Credit
- Select "FCS Innovation Academy"
- Receive approval within approximately 10 days

STEP 2: DONATE

- Follow the instructions in your approval letter
- Submit your donation (online or by mail)
- Must be completed within 60 days of approval

STEP 3: CLAIM

- File Form IT-QED-TP2 with your Georgia tax return
- Receive your full dollar-for-dollar state tax credit



gfpe.org/tax_credit/

**A Very Special Thank You
To Our Community Partner
Innovation Academy!**



**Please Support Via Tax
Credit Eligible Donation
That Goes DIRECTLY To
Innovation Academy**

Lunch Hour Sponsor Sessions



**Microsoft SQL and
AMD
in the Era of AI
by Bob Ward**



**DBA Survival Kit:
Redgate Monitor +
Toolbelt
by Chris Reyes**



DATALOGZ

**Stop Firefighting:
Build a Semantic
Layer Ready for AI
by Anouk Gorris**

***Sponsor Session Attendees Can Win A \$50 Amazon Gift Card Each
Two Raffle Gift Cards Per Session***



Session Evaluations



SQL Saturday Atlanta, AI & BI -
Session Evaluation



SQL Saturday Atlanta Closing Ceremony

5PM In The Auditorium

Must Be Present To Win Raffle Prizes





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He / him pronouns

Microsoft Data Platform MVP

MCSE: Data Management and
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Albuquerque data platform group
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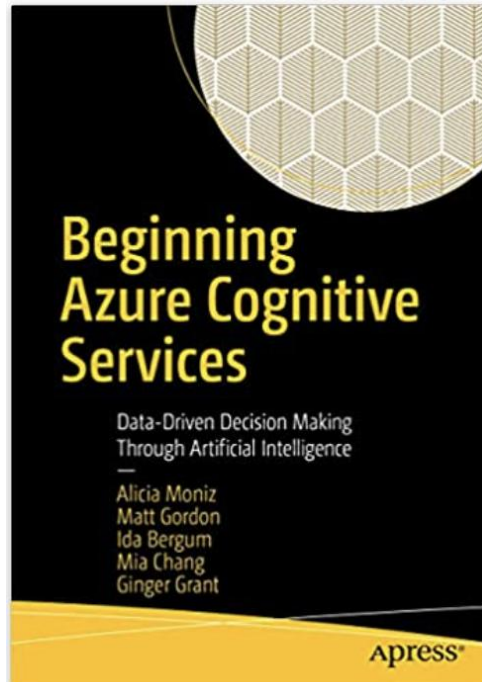
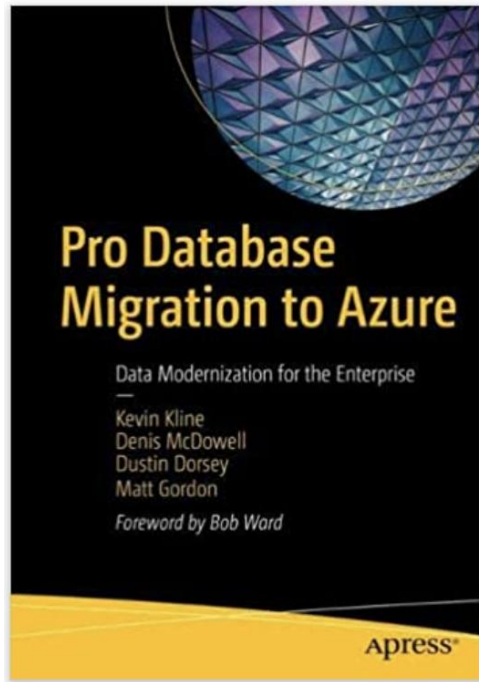
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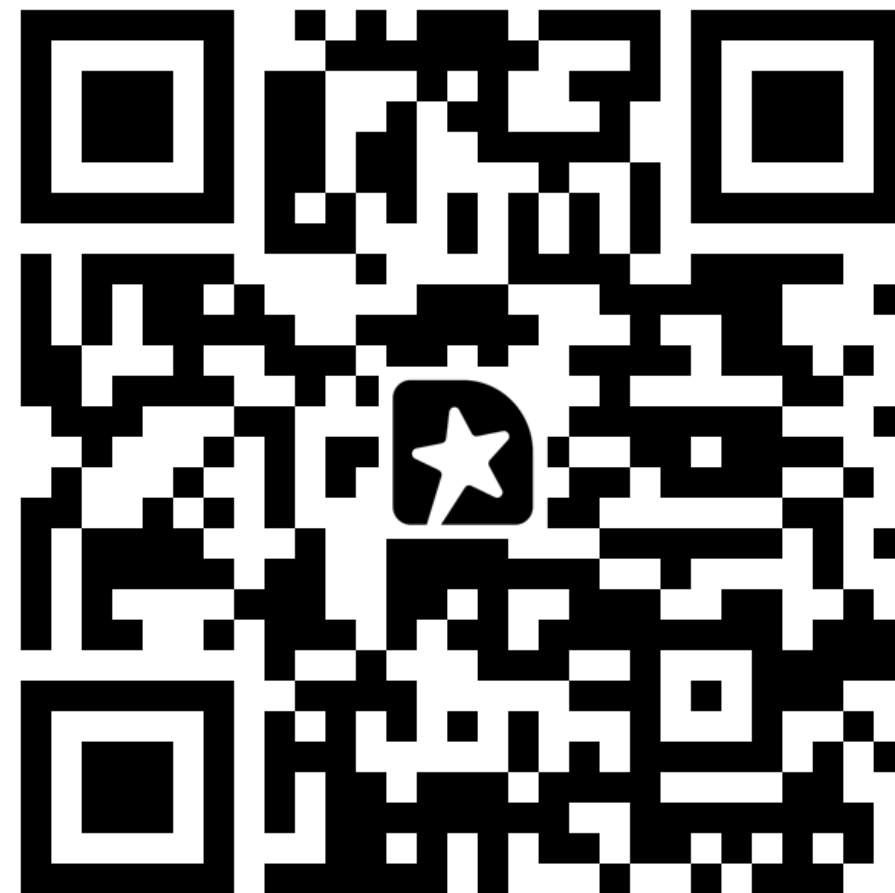
About Me



- ▶ Director, Data & Analytics for Lean TECHniques
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We Want Your
Feedback



Fabric IQ in the Real World

What Era Are We In Now?



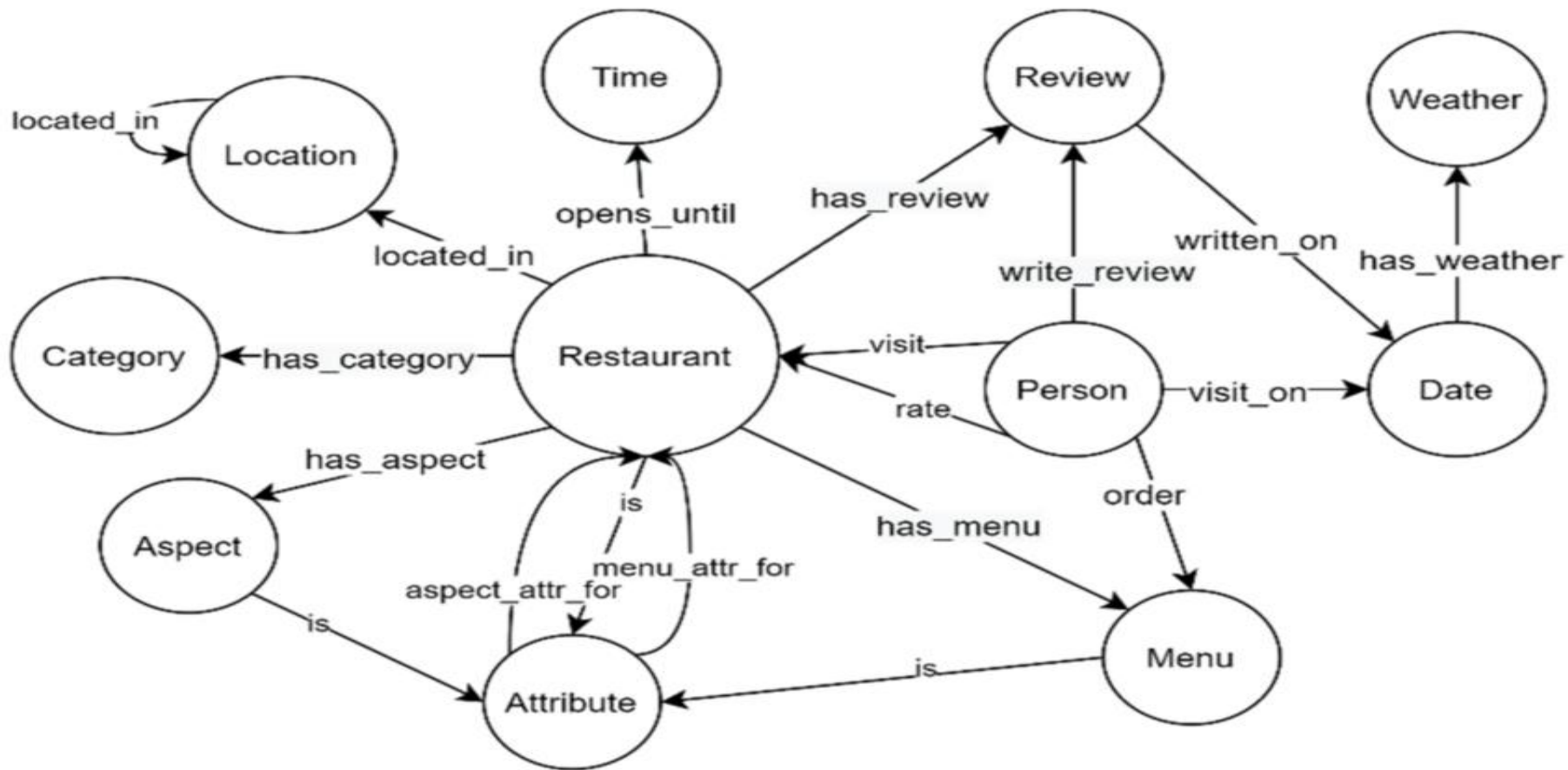
We're Nothing If Not Trendy...

- ▶ “At least 80% of governments will deploy AI agents to automate routine decision-making, enhancing efficiency and service delivery by 2028.” - *Gartner*
- ▶ “Rapid AI agent adoption will overwhelm data and analytics operations built by and for humans.” - *Gartner*
 - ▶ “Boooooo” - *humans who work in data & analytics*

Ontology explained



Applied ontology in information systems



Types of Agents in Microsoft Fabric

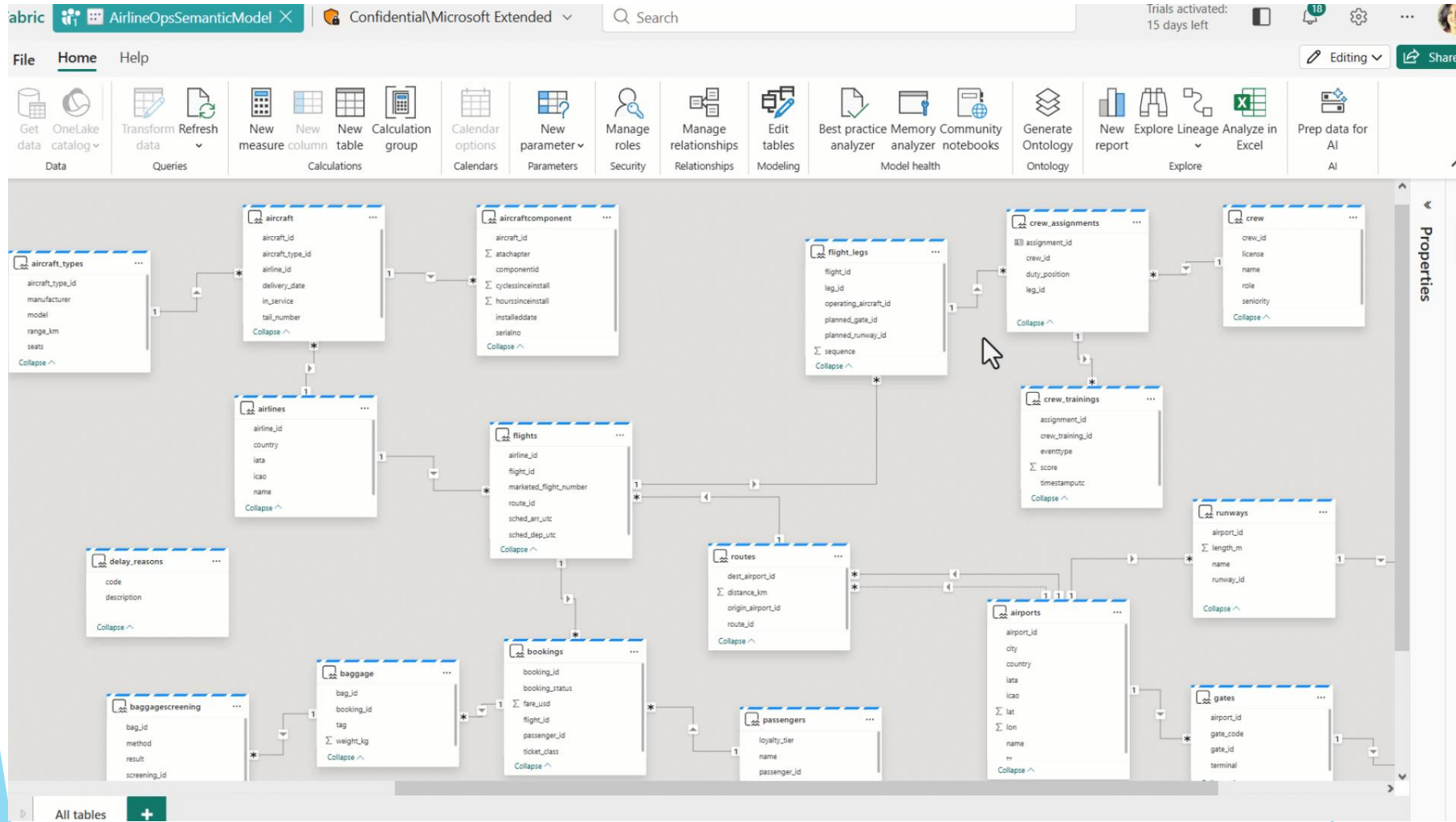
Data Agent

- ▶ You ask a question
- ▶ It translates your question into a query and gives you an answer
- ▶ Isolated investigations
- ▶ It takes no actions unless told

Operations Agent

- ▶ Can run constantly in the background
- ▶ Detects events and conditions
- ▶ It “reasons” and recommends a response
- ▶ It can take actions if permitted

Automated Ontology Generation



Sample Ontology with Enterprise Data

The screenshot displays the Microsoft Fabric interface for managing an ontology. The top navigation bar includes the Fabric logo, a workspace named 'AirlineOntology', a search bar, and a 'Trials activated: 15 days left' notification. The left sidebar contains navigation icons for Home, Workspaces, Copilot, OneLake catalog, Monitor, Real-Time, and Workloads, with 'AirlineOntology' selected.

The main content area is divided into two panels. The left panel, titled 'Ontology', shows a diagram with the following elements:

- runwayconditionreport** (top left)
- runwayconditionreport_has_...** (middle left)
- runways** (center, highlighted with a green border and a dropdown menu)
- runways_has_airports** (middle right)
- airports** (bottom right)
- flights** (bottom left)
- Flight_Has_Runway_Assigned** (bottom left, with a '+1' count)

The right panel, titled 'Entity type configuration', shows the configuration for the 'runways' entity type:

- Entity type name:** runways
- Key:** runway_id
- Instance display name:** Add instance display name
- Properties:** Bindings (selected), Report links
- Source table:** runways
- Data details:** IngiteOntologyDemoPrep / lh_airline_ops
- Binding type:** Static
- Bound properties:** airport_id, length_m, name, runway_id
- Buttons:** Add data to entity type (highlighted with a hand cursor)

Ontology Concepts/Examples

- ▶ Entity type
 - ▶ “An abstract representation of a business object. It defines the logical model of an item.”
 - ▶ Vehicle, sensor, etc.
- ▶ Relationship type
 - ▶ “A definition that specifies how two entity types are connected.”
 - ▶ Located_at, monitoring_by, etc.
- ▶ Relationship instance
 - ▶ “A specific occurrence of a relationship type between two entity instances.”
 - ▶ Many sensors -> one vehicle

Relationship Configuration Example

Home

Add entity type

Add relationship

Entity type overview

Entity Types

Ontology

Store

Product

Enter name

Store

Product

sells

Store

Relationship configuration

Source data ⓘ

Workspace *

Ontology

Lakehouse *

OntologyDataLH

Table *

factsales

Define the relationship between entity types by binding columns from your selected source to the key defined on each entity type.

1. Source entity type *

Store

Source column

StoreId

Entity type key property

StoreId

Links to

2. Target entity type *

Product

Source column

ProductId

Entity type key property

ProductId

Create

Delete relationship type

How Do We Define An Operations Agent?

Business goals
Define the business goals for the agent to accomplish. [Learn more](#)

Agent instructions
Set explicit instructions or procedures for the agent to follow.

Specify agent instructions

Bikes can be moved from station to station. Notify the operators about which station needs bikes to be allocated to. Try and keep at least 2 bikes available in any given street at all times. Tell the technicians when bike stocks are low, and which street it's low in.

If the number of available stations gets to zero, we can pick up bikes from that station to make sure arriving customers can drop off their bikes.

Knowledge source
Select a data source that the agent will use as knowledge.

bikes

KustoDatabase

+ Add data

Who Sets All This Up and Maintains It?

- ▶ Who sets up the ontology?
- ▶ Who defines it?
- ▶ Who sets up the operations agents?
- ▶ Who is allowed to modify them?

Do I Need an Ontology and a Gold Layer?

- ▶ Ontologies and gold layers are both curated
- ▶ Business processes have approved their definition and contents
- ▶ Would the same users consume both?
- ▶ Will they be consumed the same way?

How Is Ontology Relevant?

- ▶ What is the relevance for human “agents” vs. AI agents?
- ▶ Is ontology more or less relevant for SMB vs. large enterprises?
- ▶ Is defining an ontology a relevant requirement for implementing Fabric?

Who's In Charge Of My Ontology and What Will It Charge Me?

- ▶ Who should in charge of establishing and governing our ontology?
- ▶ Who should be in charge of updating our ontology?
- ▶ I only have smaller capacities (F2, F4, F8, etc.) - will this eat up some to all of my CUs?

Is Fabric IQ Integral to Fabric?

- ▶ Should we be working with Fabric IQ now or wait for it to go GA?
- ▶ Will implementing Fabric IQ benefit every user that consumes data from Fabric?
- ▶ Is Fabric IQ ready for prime time?

Guidelines and Guardrails

- ▶ Reports of high capacity usage
- ▶ Very early days for this
- ▶ Segregate this capacity from anything else

Resources

- ▶ <https://github.com/drjekyll325/presentations>
- ▶ <https://blog.fabric.microsoft.com/en-US/blog/trusted-ai-starts-with-microsoft-fabric-unified-real-time-intelligence-and-iq-context/>
- ▶ <https://learn.microsoft.com/en-us/fabric/iq/ontology/overview>
- ▶ <https://learn.microsoft.com/en-us/fabric/iq/ontology/tutorial-0-introduction?pivots=semantic-model>
- ▶ <https://learn.microsoft.com/en-us/fabric/iq/ontology/resources-capacity-usage>
- ▶ <https://sandervandeverde.wordpress.com/2025/12/25/first-look-at-fabric-ontology/>

We Want Your Feedback



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